Enhancing Multivariate Time Series Anomaly Detection With an Inference Stacked Recurrent-Autoencoder in Strong Mechanistic Contexts

Tianming Xie, Zhiwei Gao, *Fellow, IEEE*, Qifa Xu, Cuixia Jiang, and Aihua Zhang

***Abstract*—Existing self-supervised multivariate time series anomaly detection methods struggle with interference among variables during reconstruction. They also tend to miss capturing critical anomaly information, resulting in unsatisfactory performance, especially in scenarios with strong mechanistic contexts. To this end, we propose a targeted anomaly detection algorithm called inference stacked recurrent autoencoder (ISRAE). Its key contribution lies in the design of a specific inference kernel, derived from specialist knowledge, which captures the strong mechanistic relationships among variables. This kernel is then fused with the multidimensional anomalies predicted by the SRAE, which mitigates interference among variables through the stacking technique. Furthermore, a novel differential constraint is introduced into the loss function, which not only highlights anomaly reconstruction errors, but also smooths the reconstructions, enhancing overall detection performance. Comprehensive comparison experiments and ablation studies show that ISRAE achieves superior anomaly detection performance under strong mechanistic contexts and highlight the importance of each key module in ISRAE.**

***Index Terms*—Anomaly detection, inference kernel, inference stacked recurrent-autoencoder (ISRAE), multivariate time series, self-supervised method, strong mechanistic context.**

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I. Introduction

The rapid development of artificial intelligence (AI) creates opportunities to integrate AI technologies into the prognostics and health management (PHM) of industrial equipment. Anomaly detection, the initial phase of PHM, is a crucial foundation for subsequent stages. A key area within anomaly detection is multivariate time series (MTS) anomaly detection, which is becoming increasingly vital across diverse industries, including wind power, natural gas, and aerospace [1], [2]. Consequently, methods for detecting anomalies in MTS have been extensively explored to identify and locate anomalies more accurately.

According to [1] and [2], existing anomaly detection methods can be categorized into model-based [3], signal-based [4], [5], and knowledge-based methods [6], [7], [8]. Given the extreme imbalance between anomalies and normal data in real-world scenarios, along with the uncertainty of undiscovered anomalies that may deviate from known anomaly patterns, unsupervised anomaly detection methods–often referred to as self-supervised methods–are likely the most suitable choice. These methods have become the primary focus of current research and fall under the knowledge-based methods mentioned above.

The self-supervised methods primarily focus on learning typical patterns using only normal data during training. This approach ensures that the trained model accurately represents normal data while poorly representing anomalies. Consequently, anomalies can be identified through differences in representation. The more pronounced these differences are, the more accurate the anomaly detection results will be.

Among these self-supervised methods, autoencoders have become mainstream since they were first introduced by [9]. Variants of autoencoders [10], [11] detect anomalies in images or 2-D data through reconstruction. However, due to the lack of consideration for temporal dependencies, autoencoders based on fully connected or convolutional architectures perform poorly in MTS anomaly detection. The performance of models like the deep autoencoding Gaussian mixture model (DAGMM) [12] on MTS is consistently inferior to self-supervised models that consider temporal dependencies [13], [14], [15]. To address temporal dependencies, recurrent neural networks like long short-term memory (LSTM) [16] and gate recurrent unit (GRU) [17] have been integrated into autoencoders. For example, [18] first integrate LSTM into an autoencoder, achieving good reconstruction performance. Variants such as LSTM-based variational autoencoders [19], [20] and GRU-based models [13], which employ planar normalizing flow to handle temporal dependencies of latent variables, fit temporal trends better and result in more robust models. In addition to temporal dependencies, the correlation between variables has also gained attention. The authors in [14] used specific correlation matrices to describe and interpret detected anomalies, while [15] and [21] represented variable correlations through graph construction. However, excessive focus on variable correlations can increase interference and weaken the reconstruction capability of self-supervised models. This is particularly evident in existing models that consider the entire multivariate input collectively and share weights, especially when variables are relatively independent and exhibit significantly different trends. In many cases, variables of MTS do not exhibit strong correlations and are mostly independent. The causal or logical relationships between these variables only become apparent during anomalies. Consequently, the model cannot effectively learn these relationships from the limited anomaly data. To enhance reconstruction capability, [22] and [23] incorporated adversarial strategies from generative adversarial networks into autoencoders. Yet, these models show weaker generalization when the normal training data does not cover the entire distribution, performing well only within the training set distribution. Given transformers' powerful sequence processing capabilities, techniques related to transformers have been applied to MTS anomaly detection [24], [25], [26]. However, their successful application relies on massive data and a unified data representation. These requirements are challenging for MTS, especially in industrial settings with multisensor data. Multisensor data is often sensitive and proprietary, and data from different devices exhibits distributional differences. In addition, unlike transformer-based methods, there is no standardized method for normalizing such multisensor data uniformly currently.

In addition to the limitations regarding correlation and transformers, there are still gaps in current research on self-supervised MTS anomaly detection. Existing research often treats all variables in MTS equally, aggregating reconstruction errors into total errors for anomaly detection. Methods of this research can only detect obvious anomalies or those synchronized across variables. However, it may not perform well under strong mechanistic contexts, where variables in MTS often have different importances or strong logical relationships. Existing self-supervised methods trained solely on normal data cannot learn these relationships; they must be inferred from fault cases combined with mechanical principles. Therefore, incorporating mechanism-based inference is necessary to enhance anomaly detection performance in such contexts. Furthermore, [27] critically discussed the evaluation methods used for anomaly results in popular studies, including [13] and [14].

To this end, we propose a targeted anomaly detection algorithm called inference stacked recurrent autoencoder (ISRAE). The contributions of this study are summarized as follows.

1) We propose the ISRAE anomaly detection algorithm for identifying anomalies within strong mechanistic contexts. This algorithm incorporates a specialized inference kernel that detects anomalies by considering causal or logical relationships among variables.
2) ISRAE integrates various recurrent neural networks (RNNs), including LSTM and GRU, and achieves enhanced data representation through a stacking technique that effectively mitigates interference among variables.
3) We introduce a new differential constraint in the loss function, which helps the model better approximate normal data and enhance the distinction between normal data and anomalies in reconstruction errors.

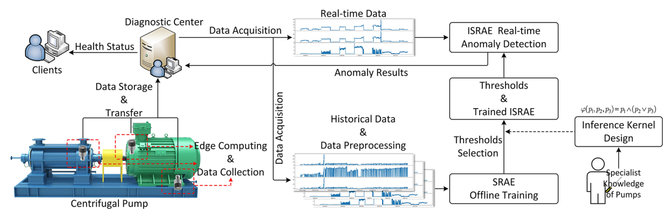

Fig. 1. Process of anomaly detection based on ISRAE.

II. Theories of ISRAE

A. Self-Supervised Based Anomaly Detection

A self-supervised method can be simply defined as $\hat{\boldsymbol{x}} = f(\boldsymbol{x})$, where the objective is to define a model that can reconstruct an output as similar to the input as possible. This indicates that if we want to use $f(.)$ for anomaly detection, the input $\boldsymbol{x}$ should be normal data so that the model can learn the normal behavior patterns of $\boldsymbol{x}$. After model $f(.)$ is trained, the reconstruction errors $\boldsymbol{e} = \boldsymbol{x} - \hat{\boldsymbol{x}}$ are finally used to detect anomalies.

B. Overall Architecture

1) Scheme: Fig. 1 illustrates the complete process of anomaly detection based on ISRAE from data collection to final detection. Initially, the vibration data is obtained from the wireless sensor and then sent to the diagnostic center for data storage. Historical data is then retrieved from the database for offline training of stacked recurrent autoencoder (SRAE). Subsequently, an inference kernel is designed based on specialist knowledge and integrated with the trained SRAE to create ISRAE. The anomaly detection threshold is determined by using test data through a traversal method. Once the threshold is set, real-time anomaly detection is conducted with the trained ISRAE model, the selected threshold, and the inference kernel. Finally, detection results are evaluated and sent back to the diagnostic center to generate the machines' health status, which is then communicated to clients. As the historical data accumulates, the model requires periodic retraining, and the thresholds need regular updates.

C. The Architecture of SRAE

SRAE must be constructed before ISRAE is created. Fig. 2 shows the architecture of SRAE. We first standardize the original data, $\mathbf{X}$. Then, we segment the data using a sliding window W (with a window length of 10), separating the MTS into three sets of single time series based on their dimensions.

As mentioned in Section I, a weight-sharing autoencoder may encounter interference among variables when reconstructing MTS. To mitigate this issue, we construct a recurrent autoencoder (RAE) for each set of single time series. Thus, we build

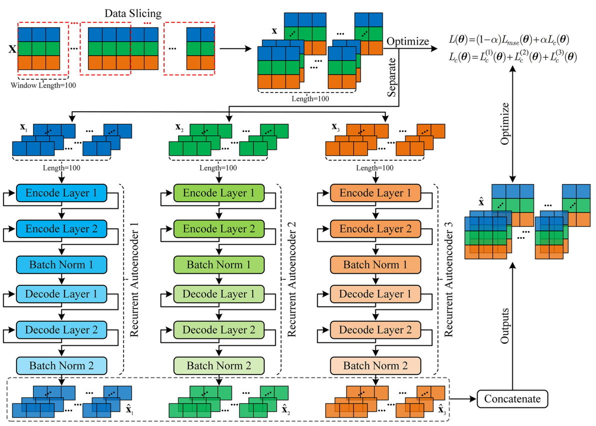

Fig. 2. Architecture of SRAE.

three RAEs and stack them together for joint training and optimization, yielding the stacked model as SRAE. The kth encoder of an RAE can be presented as follows:

$$h_{k,t} = \begin{cases} E_{k,l}\left(x_{k,t}, h_{k,t-1}^{(l)}\right), & l \leqslant 1 \\ E_{k,l}\left(h_{k,t}^{(l-1)}, h_{k,t-1}^{(l)}\right), & l > 1 \end{cases} \tag{1}$$

where $h_{k,t}$is the output of the k-th encoder; $E_{k,l}(.)$ denotes the lth layer of the kth encoder;t represents the time step in time series $\boldsymbol{x}_k$; $x_{k,t}$ is the kth dimension input of $\mathbf{X}$; $h_{k,t}^{(l)}$ refers to the hidden state of the lth ($l = 1, 2, \ldots, 4$) layer. After constructing the encoder, the subsequent decoder can be similarly written as

$$h_{k,t}^{(l')} = \begin{cases} D_{k,l'}\left(BN(h_{k,t}), h_{k,t-1}^{(l')}\right), & l' \leqslant 1 \\ D_{k,l'}\left(h_{k,t}^{(l-1)}, h_{k,t-1}^{(l')}\right), & l' > 1 \end{cases} \tag{2}$$

where $BN(.)$ is the batch normalization operation to stabilize and accelerate training; $h_{k,t}^{(l')}$ refers to the hidden state of l'th ($l' = 1, 2, \ldots, 4$) layer of the decoder. After the hidden state output $h_{k,t}^{(l')}$ from the last layer of the decoder are obtained, the final output $\hat{\boldsymbol{x}}_t$ can be expressed as

$$\hat{\boldsymbol{x}}_t = \text{concat}\left(BN\left(h_{1,t}^{(l')}\right), BN\left(h_{2,t}^{(l')}\right), BN\left(h_{3,t}^{(l')}\right)\right) \tag{3}$$

where concat(.) is the concatenation operation. Equation (3) represents the output of our model obtained by stacking outputs of multiple RAEs, which is the output of SRAE. Taking (1)–(3) together, let's denote the kth RAE as $f_k(x_{k,t})$ and SRAE as $f(\boldsymbol{x}_t)$. In this study, we let $k = 1, 2, \ldots, K$ and $K = 3$. Ultimately, SRAE can be briefly expressed as

$$f(\boldsymbol{x}_t) = \text{concat}(f_1(x_{1,t}), f_2(x_{2,t}), \ldots, f_k(x_{k,t}), \ldots, f_K(x_{K,t})). \tag{4}$$

D. Loss Function

The loss function of the SRAE model is defined as

$$L(\boldsymbol{\theta}) = \frac{(1-\alpha)}{NK}\sum_{n=1}^{N}\sum_{k=1}^{K} L_{mse}^{(k)}(\boldsymbol{\theta}) + \frac{\alpha}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} L_{C}^{(k)}(\boldsymbol{\theta}) \tag{5}$$

where α is a tunable parameter that balances the trade-off between the two terms; N denotes the batch size. The first term $\frac{(1-\alpha)}{NK}\sum_{n=1}^{N}\sum_{k=1}^{K} L_{mse}^{(k)}(\boldsymbol{\theta})$ is the average of the MSE loss across K RAE models. The second term $\frac{\alpha}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} L_{C}^{(k)}(\boldsymbol{\theta})$ is a constraint we propose to minimize the influence of unlabeled anomalies or individual noise in the training set. We do not average the second term because averaging may cause interference between variables during reconstruction, making the constraint effective only within individual variables. Instead, we sum up all three constraints. This constraint aims to smooth the reconstruction results by minimizing the differences between consecutive points. It helps the model better capture the patterns of normal data and reconstruct data that closely resembles normal instances even when anomalies are present. As a result, the final reconstruction errors can more effectively differentiate between normal data and real anomalies. The constraint $L_C^{(k)}(\boldsymbol{\theta})$ in (5) can be written as

$$L_C^{(k)}(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T}(x_{k,t+1} - x_{k,t})^2. \tag{6}$$

E. Convergence of SRAE's Training

SRAE's ability to capture the normal behavior patterns of MTS data is critical to the anomaly detection performance of the final ISRAE model, with its convergence during training crucial for learning these patterns effectively. In (5), both $L_{mse}^{(k)}(\boldsymbol{\theta})$ and $L_C^{(k)}(\boldsymbol{\theta})$ take similar forms of squared difference, which is easier to satisfy the requirement of convexity compared to the entire loss function $L(\boldsymbol{\theta})$. This convexity is crucial for the subsequent proof. If we can prove the convergence of $L_{mse}^{(k)}(\boldsymbol{\theta})$ during training, the convergence of $L_C^{(k)}(\boldsymbol{\theta})$ can similarly be established due to their identical form. Ultimately, based on the addition rule of inequalities, we can infer that $L(\boldsymbol{\theta})$ converges.

Before the theoretical proof, two assumptions are needed. For the SRAE model, consider any two parameters, $\boldsymbol{\theta}_{k,1}$ and $\boldsymbol{\theta}_{k,2}$, obtained from different iterations during training. Here, $k \in K$ represents the kth dimension of the model's parameters. According to [28], these two parameters should meet the following conditions:

$$\mathbf{A1}: \quad \|\theta_{k,1} - \theta_{k,2}\|_2 \leqslant d_k \tag{7}$$

where d represents the distance boundary. Since $\theta_{k,1}$ and $\theta_{k,2}$ are obtained through the Adam optimization algorithm, **A1** holds for any differentiable function. Another assumption is the boundedness of gradients for any iteration, $\forall i$, during the training process. For the ith iteration, the gradient $g_{k,i}$ satisfies

$$\mathbf{A2}: \quad \|g_{k,i}\|_2 \leqslant d'_k. \tag{8}$$

We use learning rate decay during model training to satisfy the first assumption as much as possible. In addition, to meet the second assumption, we use gradient clipping, Xavier initialization, and batch normalization. We measure the convergence by introducing the regret conception from [29], with the form

$$C(N) = \frac{1}{K}\sum_{k=1}^{K}\sum_{i=1}^{N}\left(L_i^{(k)}(\theta_i) - L^{(k)}(\theta^*)\right) \tag{9}$$

where $\boldsymbol{\theta}_i$ denotes the parameter set of SRAE from the ith iteration during training, and $\boldsymbol{\theta}^*$ represents the parameter set that optimizes the loss function $L(.)$. When $N \to \infty$, if $C(N)/N \to 0$, it implies that $L_i(\boldsymbol{\theta}_i)$ is approaching $L_i(\boldsymbol{\theta}^*)$, indicating that the loss function $L(.)$ converges during training. Therefore, $\lim\limits_{N\to\infty} \frac{C(N)}{N} = 0$ serves as the criterion for determining the convergence of the loss function $L(.)$.

We train the model using the Adam optimizer [28] and have

$$\begin{aligned}\boldsymbol{m}_i &= \beta_1 \boldsymbol{m}_{i-1} + (1-\beta_1)\,\boldsymbol{g}_i\\ \boldsymbol{v}_i &= \beta_2 \boldsymbol{v}_{i-1} + (1-\beta_2)\,\boldsymbol{g}_i^2\\ \hat{\boldsymbol{m}}_i &= \frac{\boldsymbol{m}_i}{1-\beta_1^i}\\ \hat{\boldsymbol{v}}_i &= \frac{\boldsymbol{v}_i}{1-\beta_2^i}\\ \boldsymbol{\theta}_{i+1} &= \boldsymbol{\theta}_i - \frac{\eta}{\sqrt{\hat{\boldsymbol{v}}_i + \boldsymbol{\varepsilon}}}\hat{\boldsymbol{m}}_i. \end{aligned} \tag{10}$$

To establish the direct relationship between $\boldsymbol{m}_i$ and $\boldsymbol{g}_i$, we expand the recursive expression $\boldsymbol{m}_i = \beta_1 \boldsymbol{m}_{i-1} + (1-\beta_1)\boldsymbol{g}_i$ in (10) as

$$\boldsymbol{m}_i = (1-\beta_1)\sum_{j=1}^{i} \beta_1^{i-j}\boldsymbol{g}_j. \tag{11}$$

Similarly, by expanding $\boldsymbol{v}_i = \beta_2 \boldsymbol{v}_{i-1} + (1-\beta_2)\boldsymbol{g}_i^2$, we have

$$\boldsymbol{v}_i = (1-\beta_2)\sum_{j=1}^{i} \beta_2^{i-j}\boldsymbol{g}_j^2. \tag{12}$$

For the ith iteration in the local optimum region that ensures convexity, there exists

$$L_i(\boldsymbol{\theta}_i) - L_i(\boldsymbol{\theta}^*) \leqslant \langle \boldsymbol{g}_i, \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle\,. \tag{13}$$

According to (9) and (13), we derive the following inequality:

$$C(N) \leqslant \sum_{i=1}^{N} \langle \boldsymbol{g}_i, \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle \tag{14}$$

where $\langle \boldsymbol{g}_i, \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle$ can also be represented as $\langle \boldsymbol{g}_i, \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle = \sum_{k=1}^{K} g_{k,i}(\theta_{k,i} - \theta_k^*)$. Therefore, (14) can be ultimately expressed as

$$C(N) \leqslant \sum_{i=1}^{N}\sum_{k=1}^{K} g_{k,i}(\theta_{k,i} - \theta_k^*). \tag{15}$$

Therefore, our objective shifts to proving the boundedness of $\sum_{k=1}^{K} g_{k,i}(\theta_{k,i} - \theta_k^*)$. According to (10)–(12), for $\theta_{k,i}$, which represents the kth dimension of the parameters $\boldsymbol{\theta}$ after the ith training iteration, we have

$$\theta_{k,i+1} = \theta_{k,i} - \eta_i \frac{1}{1-\beta_1^i}\frac{\beta_1 m_{k,i-1} + (1-\beta_1)\,g_{k,i}}{\sqrt{\hat{v}_{k,i}}} \tag{16}$$

As β_1 denotes the decay rate, let us assume it is monotonically nondecreasing. As the iteration increases, (16) can be further expressed as

$$\theta_{k,i+1} = \theta_{k,i} - \eta_i \frac{1}{1-\prod_{j=1}^{i}\beta_{1,j}}\frac{\beta_{1,i} m_{i-1} + (1-\beta_{1,i}) g_{k,i}}{\sqrt{\hat{v}_{k,i}}}. \tag{17}$$

Let $\lambda_i = \eta_i \frac{1}{1-\prod_{j=1}^{i}\beta_{1,j}}$, then (17) can be written as

$$\theta_{k,i+1} = \theta_{k,i} - \lambda_i \frac{\beta_{1,i} m_{k,i-1} + (1-\beta_{1,i}) g_{k,i}}{\sqrt{\hat{v}_{k,i}}}. \tag{18}$$

Since we try to extract $g_{k,i}(\theta_{k,i} - \theta_k^*)$, (18) can be transformed as

$$\begin{aligned}&\theta_{k,i+1} = \theta_{k,i} - \lambda_i \frac{\beta_{1,i} m_{k,i-1} + (1-\beta_{1,i}) g_{k,i}}{\sqrt{\hat{v}_{k,i}}}\\ &\Rightarrow 2\lambda_i \frac{m_{k,i}}{\sqrt{\hat{v}_{k,i}}}(\theta_{k,i}-\theta_k^*) = (\theta_{k,i}-\theta_k^*)^2 - (\theta_{k,i+1}-\theta_k^*)^2 + \lambda_i^2 \frac{m_{k,i}^2}{\hat{v}_{k.i}}. \end{aligned} \tag{19}$$

By introducing $g_{k,i}$ on both sides of (19), we can extract

$$\begin{aligned} g_{k,i}(\theta_{k,i} - \theta_k^*) &= \underbrace{\frac{\sqrt{\hat{v}_{k,i}}\left[(\theta_{k,i}-\theta_k^*)^2 - (\theta_{k,i+1}-\theta_k^*)^2\right]}{2\lambda_i(1-\beta_{1,i})}}_{(1)}\\ &\quad - \underbrace{\frac{\beta_{1,i}}{1-\beta_{1,i}} m_{k,i-1}(\theta_{k,i}-\theta_k^*)}_{(2)} + \underbrace{\frac{\lambda_i}{2(1-\beta_{1,i})}\frac{m_{k,i}^2}{\sqrt{\hat{v}_{k,i}}}}_{(3)}. \end{aligned} \tag{20}$$

According to (20), $\sum_{k=1}^{K} g_{k,i}(\theta_{k,i} - \theta_k^*)$ can be expressed as

$$\begin{aligned} &\sum_{i=1}^{N} g_{k,i}(\theta_{k,i} - \theta_k^*)\\ &= \sum_{i=1}^{N}\left(\underbrace{\frac{\sqrt{\hat{v}_{k,i}}\left[(\theta_{k,i}-\theta_k^*)^2 - (\theta_{k,i+1}-\theta_k^*)^2\right]}{2\lambda_i(1-\beta_{1,i})}}_{(1)}\right.\\ &\quad \left. - \underbrace{\frac{\beta_{1,i}}{1-\beta_{1,i}} m_{k,i-1}(\theta_{k,i}-\theta_k^*)}_{(2)} + \underbrace{\frac{\lambda_i}{2(1-\beta_{1,i})}\frac{m_{k,i}^2}{\sqrt{\hat{v}_{k,i}}}}_{(3)}\right). \end{aligned} \tag{21}$$

Let us scale terms (1), (2), and (3) in (21). In term (1), let us expand λ_i. Furthermore, based on the first assumption, we can infer that $\frac{\sqrt{\hat{v}_{k,i}}(\theta_{k,i}-\theta_k^*)^2}{2\eta_i(1-\beta_{1,1})} \leqslant \frac{\sqrt{\hat{v}_{k,i}}d_k^2}{2\eta_i(1-\beta_{1,1})}$, where $-\frac{\sqrt{\hat{v}_{k,i}}(\theta_{k,i}-\theta_k^*)^2}{2\eta_i(1-\beta_{1,1})} \leqslant 0$. With all these considerations, we can derive

$$\sum_{i=1}^{N} \frac{\sqrt{\hat{v}_{k,i}}\left[(\theta_{k,i}-\theta_k^*)^2 - (\theta_{k,i+1}-\theta_k^*)^2\right]}{2\lambda_i(1-\beta_{1,i})}$$

$$\leqslant \frac{\sqrt{\hat{v}_{k,1}}(\theta_{k,1}-\theta_k^*)^2}{2\eta_1(1-\beta_{1,1})} - \frac{\sqrt{\hat{v}_{k,N}}(\theta_{k,N+1}-\theta_k^*)^2}{2\eta_N(1-\beta_{1,1})}$$
$$+\sum_{i=2}^{N}(\theta_{k,i}-\theta_k^*)^2\left[\frac{\sqrt{\hat{v}_{k,i}}}{2\eta_i(1-\beta_{1,1})}-\frac{\sqrt{\hat{v}_{k,i-1}}}{2\eta_{i-1}(1-\beta_{1,1})}\right]$$
$$\leqslant \frac{d_k^2\sqrt{\hat{v}_{k,N}}}{2\eta_N(1-\beta_{1,1})} \tag{22}$$

Given the conditions $\hat{v}_{k,i}=\frac{v_{k,i}}{1-\beta_{2,i}}$, $v_{k,i}=(1-\beta_2)\sum_{j=1}^{i}\beta_2^{i-j}g_{k,j}^2$, and assumption **A2**, we can derive the following:

$$\hat{v}_{k,i}=\frac{v_{k,i}}{1-\beta_2^i}=\frac{(1-\beta_2)\sum_{j=1}^{i}\beta_2^{i-j}g_{k,j}^2}{1-\beta_2^i}\leqslant (d_k')^2. \tag{23}$$

According to (22) and (23), term (1) can be scaled to

$$\frac{\sqrt{\hat{v}_{k,i}}\left[(\theta_{k,i}-\theta_k^*)^2-(\theta_{k,i+1}-\theta_k^*)^2\right]}{2\lambda_i(1-\beta_{1,i})}\leqslant\frac{d_k^2 d_k'}{2\eta_N(1-\beta_{1,1})}. \tag{24}$$

For term (2), based on the first assumption, we have

$$\frac{\beta_{1,i}}{1-\beta_{1,i}}m_{k,i-1}(\theta_{k,i}-\theta_k^*)\leqslant\frac{\beta_{1,i}}{1-\beta_{1,i}}m_{k,i-1}d_k. \tag{25}$$

As β_1 is monotonically nondecreasing, further expanding and merging $m_k^{(i)}$ based on (11) yields

$$m_{k,i}=\sum_{j=1}^{i}(1-\beta_{1,j})\left(\prod_{l=j+1}^{i}\beta_{1,l}\right)g_{k,j}. \tag{26}$$

Based on the second assumption, (26) can be further scaled to

$$\sum_{j=1}^{i}(1-\beta_{1,j})\left(\prod_{l=j+1}^{i}\beta_{1,l}\right)g_{k,j}\leqslant d_k'\left(1-\prod_{l=1}^{i}\beta_{1,l}\right)\leqslant d_k'. \tag{27}$$

Combining (25)–(27), term (2) can be ultimately scaled to

$$\frac{\beta_{1,i}}{1-\beta_{1,i}}m_{k,i-1}(\theta_{k,i}-\theta_k^*)\leqslant d_k d_k'\frac{\beta_{1,i}}{1-\beta_{1,i}}. \tag{28}$$

For term (3), we mainly focus on $\frac{m_{k,i}^2}{\sqrt{\hat{v}_{k,i}}}$. According to $\hat{v}_{k,i}=\frac{v_{k,i}}{1-\beta_2^i}$, we have

$$\frac{m_{k,i}^2}{\sqrt{\hat{v}_{k,i}}}=\sqrt{1-\beta_{2,i}}\frac{m_{k,i}^2}{\sqrt{v_{k,i}}}\leqslant\frac{m_{k,i}^2}{\sqrt{v_{k,i}}}. \tag{29}$$

Let us transform $m_{k,i}^2$ based on $m_{k,i}=\sum_{j=1}^{i}(1-\beta_{1,j})(\prod_{l=j+1}^{i}\beta_{1,l})g_{k,j}$. Then, we obtain

$$m_{k,i}^2=\sum_{j=1}^{i}\frac{(1-\beta_{1,j})^2\left(\prod_{l=j+1}^{i}\beta_{1,l}\right)^2}{(1-\beta_2)\beta_2^{i-j}}\cdot v_{k,i} \tag{30}$$

According to (23) and (30), term (3) can be scaled to

$$\frac{\lambda_i}{2(1-\beta_{1.i})}\frac{m_{k,i}^2}{\sqrt{\hat{v}_{k,i}}}\leqslant\frac{\lambda_i}{2(1-\beta_{1.i})}\sum_{j=1}^{i}\frac{(1-\beta_{1,j})^2\left(\prod_{l=j+1}^{i}\beta_{1,l}\right)^2}{(1-\beta_2)\beta_2^{i-j}}\cdot d_k'. \tag{31}$$

Taking (15), (21), (24), (28), and (31) together, we can ultimately derive

$$C(N)\leqslant\sum_{i=1}^{N}\sum_{k=1}^{K}\left(\frac{d_k^2 d_k'}{2\eta_N(1-\beta_{1,1})}+d_k d_k'\frac{\beta_{1,i}}{1-\beta_{1,i}}\right.$$
$$\left.+\frac{\lambda_i}{2(1-\beta_{1,i})}\sum_{j=1}^{i}\frac{(1-\beta_{1,j})^2\left(\prod_{l=j+1}^{i}\beta_{1,l}\right)^2}{(1-\beta_2)\beta_2^{i-j}}\cdot d_k'\right) \tag{32}$$

Since all the values on the right side of the inequality in (32) stay constant, we infer that as $N\to\infty$, $\lim_{N\to\infty}\frac{C(N)}{N}=0$. Thus, we can prove that $L_{mse}^{(m)}(\boldsymbol{\theta})$ converges during training. Similarly, convergence of $L_C^{(m)}(\boldsymbol{\theta})$ can also be proved. Finally, following the rule of inequalities, we conclude that the loss function $L(\boldsymbol{\theta})$, represented by (5), can converge during the model's training.

Since the convergence of ISRAE during the training process has been proved we train the model using the Adam algorithm to optimize

$$\boldsymbol{\theta}^*=\arg\min_{\boldsymbol{\theta}}\left(\frac{(1-\alpha)}{NM}\sum_{n=1}^{N}\sum_{k=1}^{K}L_{mse}^{(k)}(\boldsymbol{\theta})+\frac{\alpha}{N}\sum_{n=1}^{N}\sum_{k=1}^{K}L_C^{(k)}(\boldsymbol{\theta})\right). \tag{33}$$

F. Anomaly Detection Algorithm Based on ISRAE

After training the SRAE, we propose an inference kernel based on specialist knowledge for the anomaly detection phase. This kernel helps focus on detecting anomalies related to the machine's rotational frequency, such as unbalanced bearings and misalignments. These anomalies often show significant increases in energy at one-time or two-time rotational frequencies and in the RMS. Therefore, we define the inference kernel as

$$\varphi(p_1,p_2,p_3)=p_1\wedge(p_2\vee p_3) \tag{34}$$

where p_1, p_2, p_3 are the predictions of three dimensions. By integrating the inference kernel with SRAE, the final ISRAE anomaly detection algorithm is detailed in Algorithm 1.

III. Experimental Design and Results

A. Experimental Design

1) Comparative Models: We compare ISRAE with nine state-of-the-art (SOTA) models and two baselines using a private dataset, the MSPD to demonstrate the superiority of our model. The detailed descriptions of these SOTAs and baselines are presented in Table I.

Algorithm 1: ISRAE Anomaly Detection Algorithm.

Input: The SRAE model $f(\boldsymbol{x})$, training set $D_{\boldsymbol{x}} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_t, \ldots, \boldsymbol{x}_T\} \in \mathbb{R}^{T\times K\times W}$, test set $D_{\boldsymbol{x}'} = \{\boldsymbol{x}_1^{'}, \boldsymbol{x}_2^{'}, \ldots, \boldsymbol{x}_t^{'}, \ldots, \boldsymbol{x}_T^{'}\} \in \mathbb{R}^{T\times K\times W}$, test label $D_y = \{y_1, y_2, \ldots, y_t, \ldots, y_T\}$, the real-time data $\boldsymbol{x}_r$, the threshold set $D_\nu = \{\nu_1, \nu_2, \cdots, \nu_l, \cdots, \nu_L\}$, the inference kernel $\varphi(p_1, p_2, p_3)$, the initialized optimal F1-score of k-th variable $f_1^{(k)*}$; the initialized optimal threshold ν_k^*
Output: Predicted result $\hat{y}_r$

1 $\alpha \leftarrow$ Select the α in **Eq.** (5) with grid search method
2 $f(\boldsymbol{x}) \leftarrow$ Obtain the trained SRAE model
3 $D_{\hat{\boldsymbol{x}}'} \leftarrow$ Reconstruct the test set with $f(\boldsymbol{x})$
4 $E \leftarrow$ Calculate the reconstruction errors with $D_{\boldsymbol{x}'} - D_{\hat{\boldsymbol{x}}}$
5 **for** $l \leftarrow 1$ **to** L **do**
6   **for** $k \leftarrow 1$ **to** K **do**
7         $D_{y'}^{(k)} \leftarrow$ Predict anomaly with threshold $\nu_l^{(k)}$
8         $f_1^{(k)} \leftarrow$ Calculate F1-score with D_y and $D_{y'}^{(k)}$
9   **if** $f_1^{(k)} \geqslant f_1^{(k)*}$ **then**
10             $f_1^{(k)*} \leftarrow f_1^{(k)}$; $\nu_k^* \leftarrow \nu_l$
11   **end**
12   **end**
13 **end**
14 $\hat{\boldsymbol{x}}_r \leftarrow$ Reconstruct the real-time data $\boldsymbol{x}_r$ with trained SRAE
15 $\boldsymbol{e}_r \leftarrow$ Calculate the reconstruction errors $\boldsymbol{x}_r - \hat{\boldsymbol{x}}_r$
16 $\hat{y}_r^{(1)}, \hat{y}_r^{(2)}, \hat{y}_r^{(3)} \leftarrow$ Predict anomaly with threshold $\nu_1^*, \nu_2^*, \nu_3^*$
17 $\hat{y}_r \leftarrow$ Obtain final anomaly result with $\varphi(\hat{y}_r^{(1)}, \hat{y}_r^{(2)}, \hat{y}_r^{(3)})$
18 **return** $\hat{y}_r$

TABLE I
DESCRIPTIONS OF SOTAS AND BASELINES

Model	Model Descriptions
DAGMM [12]	Constrains the distribution of deep representations using a Gaussian mixture model (GMM) within a conventional autoencoder.
USAD [22]	Incorporates an adversarial learning strategy into a traditional autoencoder.
EncDec-AD [18]	Utilize LSTM to reconstruct time series, capturing temporal patterns.
LSTM-VAE [19]	Integrates variational inference methods into LSTM to enhance model robustness.
MSCRED [14]	Designs signature matrices to represent correlations between variables and uses ConvLSTM to capture spatiotemporal patterns.
OmniAnomaly [13]	Integrates variational inference into a GRU autoencoder and combines it with planar normalizing flows to capture complex distributions.
GDN [30]	Employs graph-structured learning within neural networks to capture multi-sensor relationships.
MTAD-GAT [15]	Uses graph attention networks to learn complex dependencies in both temporal and variable dimensions.
TranAD [24]	Integrates transformer architecture to enhance the model's ability to handle time series.
NAE [27]	Random noise is used as reconstruction errors to generate anomaly results.
DAE [27]	Employs original data for reconstruction errors to get anomaly results.

2) Threshold Selection: A uniform selection strategy for the predicted threshold is utilized for each model to ensure fair comparison. We generate numerous thresholds covering the range of reconstruction errors and choose the one that best detects anomalies in the test set. The details of the threshold selection are outlined in steps (4)–(12) of Algorithm 1.

3) Evaluation Metrics: Existing studies usually perform point adjustments (PA) on the predictions results. According to [27], their PA can be written as follows:

$$y_t' = \begin{cases} 1, \text{ if } e_t \geqslant \nu^* \text{or } t \in S_i \text{ and } \underset{t'\in S_i}{\exists}\ e_{t'} \geqslant \nu^* \\ 0, \text{ otherwise} \end{cases} \tag{35}$$

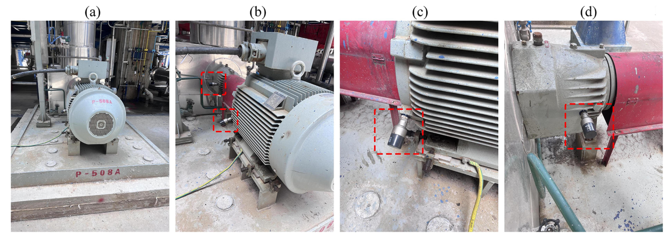

Fig. 3. Installation of sensors on a pump. (a) Overview of machine. (b) Overview of sensor installation. (c) Sensor on motor drive end. (d) Sensor on pump drive end.

TABLE II
DETAILED INFORMATION ON THE MSPD

Property	Training set	Test set	Total
Number of entities	71	71	71
Number of dimensions	3	3	3
Dataset size	141 630	95 040	236 670
Fault ratio (%)	0	3.16	1.27

where S_i denotes the ith segment of anomaly. To comprehensively evaluate the anomaly detection results, we further utilize a new PA adjustment method of [27] as follows:

$$y_t' = \begin{cases} 1, \text{ if } e_t \geqslant \nu^* \text{or } t \in S_i \text{ and } \frac{|\{t'|t'\in S_i, e_{t'} \geqslant \nu^*\}|}{|S_i|} \geqslant \tau \\ 0, \text{ otherwise.} \end{cases} \tag{36}$$

Equation (36) indicates that an anomaly segment can only be correctly predicted if more than $\tau \times |S_i|$ anomalies within that segment are accurately identified.

4) Experiment Setting: The proposed ISRAE are implemented in a PyTorch 1.8.1 environment and run on an 11th Gen Intel(R) (6-Core 2.50 GHz) processor with 64GB RAM and an NVIDIA GeForce RTX 3080.

B. Experimental Results

1) Data: The existing MTS public datasets do not contain the specialist knowledge mentioned in this study. Moreover, extracting such knowledge directly from public datasets is challenging without sufficient domain expertise. Therefore, we use a private dataset, the MSPD, to comprehensively demonstrate the superiority of ISRAE.

The MSPD is a 3-D MTS obtained from multiple sensors installed on a pump. The sensor installation is illustrated in Fig. 3, and detailed information on the MSPD is presented in Table II. The time interval between two consecutive data points is two hours in both the training and test sets. Fig. 4 shows the data collected from four specific pumps. Metrics 1–3 correspond to the RMS of velocity, the energy at the one-time rotational frequency, and the energy at the two-time rotational frequency, respectively. In this context, the one-time rotational frequency indicates the rotational frequency. Data visualization indicates a genuine anomaly when metric 1 significantly rises and either metric 2 or metric 3 also shows a substantial increase. In other cases, even if metrics 2 and 3 simultaneously rise significantly, it does not necessarily indicate genuine anomalies. These characteristics are supported by robust mechanical theories or specialist knowledge, constituting the strong mechanistic contexts mentioned before.

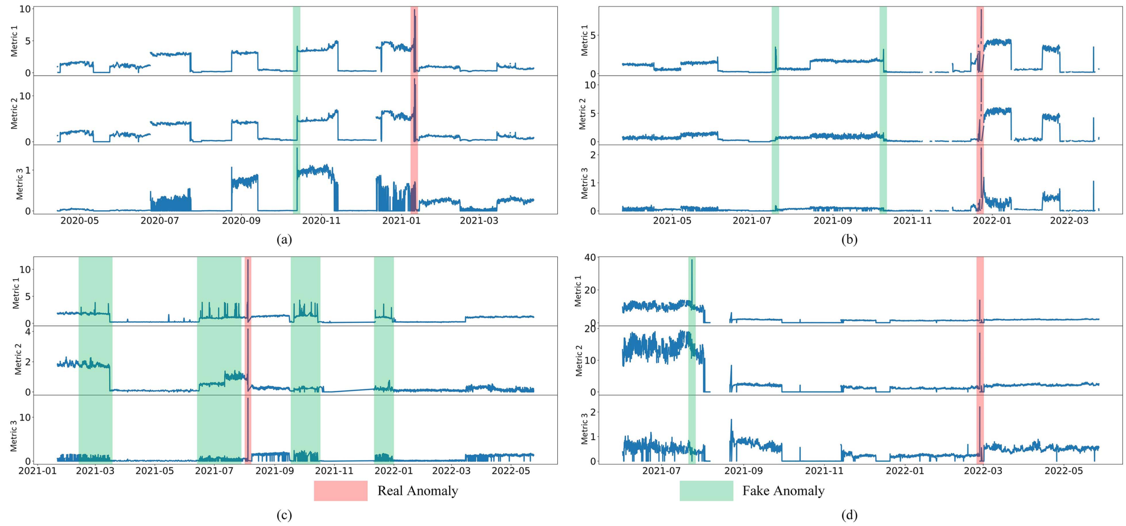

Fig. 4. MSPD visualization. (a) Pump 1. (b) Pump 2. (c) Pump 3. (d) Pump 4.

TABLE III
COMPARISON RESULTS ON THE MSPD

Model	MSPD P	MSPD R	MSPD F1	MSPD (SNR=10dB) P	MSPD (SNR=10dB) R	MSPD (SNR=10dB) F1
DAGMM	.7842 (.0051)	.8588 (.0110)	.8212 (.0069)	.7436 (.0098)	.7926 (.0186)	.7672 (.0127)
USAD	.2044 (.0066)	.4400 (.0083)	.2781 (.0066)	.2733 (.0118)	.4389 (.0091)	.3368 (.0103)
EncDec-AD	.8588 (.0059)	.8908 (.0077)	.8743 (.0019)	.7539 (.0165)	.8132 (.0209)	.7819 (.0249)
LSTM-VAE	.8716 (.0245)	.8869 (.0219)	.8789 (.0091)	.8501 (.0311)	**.8672 (.0229)**	.8586 (.0259)
MSCRED	.3904 (.0557)	.5111 (.0818)	.4345 (.0220)	.4233 (.0364)	.4978 (.0289)	.4578 (.0355)
OmniAnomaly	.9161 (.0489)	.7309 (.1069)	.8059 (.0625)	.8812 (.0326)	.7033 (.0931)	.7825 (.0631)
GDN	.7824 (.0075)	**.9050 (.0053)**	.8391 (.0065)	.7119 (.0241)	.8267 (.0184)	.7656 (.0233)
MTAD-GAT	.7859 (.0068)	.8611 (.0052)	.8217 (.0068)	.7033 (.0168)	.7916 (.0098)	.7433 (.0105)
TranAD	.7849 (.0008)	.8616 (.0026)	.8215 (.0012)	.7231 (.0101)	.8077 (.0129)	.7645 (.0121)
NAE	.2175 (.0509)	.4158 (.0812)	.2822 (.0379)	.2175 (.0509)	.4158 (.0812)	.2822 (.0379)
DAE	.8369 (.0000)	.8483 (.0000)	.8426 (.0000)	.7698 (.0000)	.8002 (.0000)	.7847 (.0000)
ISLSTMAE	.9391 (.0106)	.8531 (.0256)	**.8937 (.0109)**	.8936 (.0203)	.8421 (.0277)	**.8665 (.0213)**
ISGRUAE	**.9405 (.0106)**	.8401 (.0309)	.8870 (.0135)	**.9243 (.0156)**	.8030 (.0223)	.8579 (.0172)

Note: (1) P, R, and F1 denote precision, recall, and F1-score, respectively. (2) Boldface represents the best result; underline indicates the second-best.

2) Comparison Results: To demonstrate the effectiveness of ISRAE in detecting anomalies in the MSPD, we compare ISRAE with nine other SOTAs. To further validate the ISRAE's universality and robustness, we add noise to the MSPD at a signal-to-noise ratio (SNR) of 10 dB. Since LSTM and GRU are the two most well-established types of RNNs, we also create ISLSTMAE and ISGRUAE as two representatives of ISRAE. The comparison results reported in Table III include the average value and standard deviation across twenty repeated experiments. The F1-score, the harmonic mean of precision and recall used for threshold selection, is the primary focus during model analysis and comparison.

Table III shows that both ISLSTMAE and ISGRUAE achieve the best (F1-score = 0.8937) and second-best (F1-score = 0.8870) performance on the MSPD, respectively. We can observe that these two models achieve the best and second-best precision scores, respectively, while also maintaining relatively high recall levels. This indicates that the proposed inference kernel in ISRAE significantly reduces false alarm rates while effectively capturing most of the anomalies. USAD has the worst performance with an F1-score of only 0.2781. Its low precision of 0.2044 highlights the issue of incorrectly identifying many normal data points as anomalies. This issue may stem from overfitting in the generative adversarial learning process of USAD fusion. As a result, the model can learn the training set data perfectly but struggles to perform well on the test set. Compared to GDN and MTAD-GAT, MSCRED performs poorly because it doesn't use attention mechanisms to learn important correlations and fails to consider sufficiently long temporal dependencies like other models that integrate RNNs. This results in MSCRED having the second-worst performance on the MSPD. OmniAnomaly's F1-score is only 0.8059, significantly lower than the baseline DAE, and its recall of 0.7309 indicates that the model misses many anomalies. This suggests that OmniAnomaly does not effectively learn the patterns of normal data. Furthermore, Table III also shows that models like DAGMM, GDN, MTAD-GAT, and TranAD, while performing relatively well, cannot adequately consider the temporal dependencies of MTS due to their structural characteristics. As a result, their F1-scores are quite similar, achieving only around 0.82–0.83. For EncDec-AD and LSTM-VAE, which consider temporal dependencies, both models achieve an F1-score of 0.87, with recall and precision at relatively good levels.

In the results for the MSPD (SNR = 10 dB), ISRAE still achieves almost the best performance. ISLSTMAE and ISGRUAE have the best and third-best F1-scores, respectively. Even as the third-best, ISGRUAE's F1-score of 0.8579 is nearly identical to the second-best score of 0.8586. ISRAE's excellent performance on the noise-added MSPD dataset is due to its strong reconstruction and smoothing capabilities from the stacking method and differential constraint, and the inference

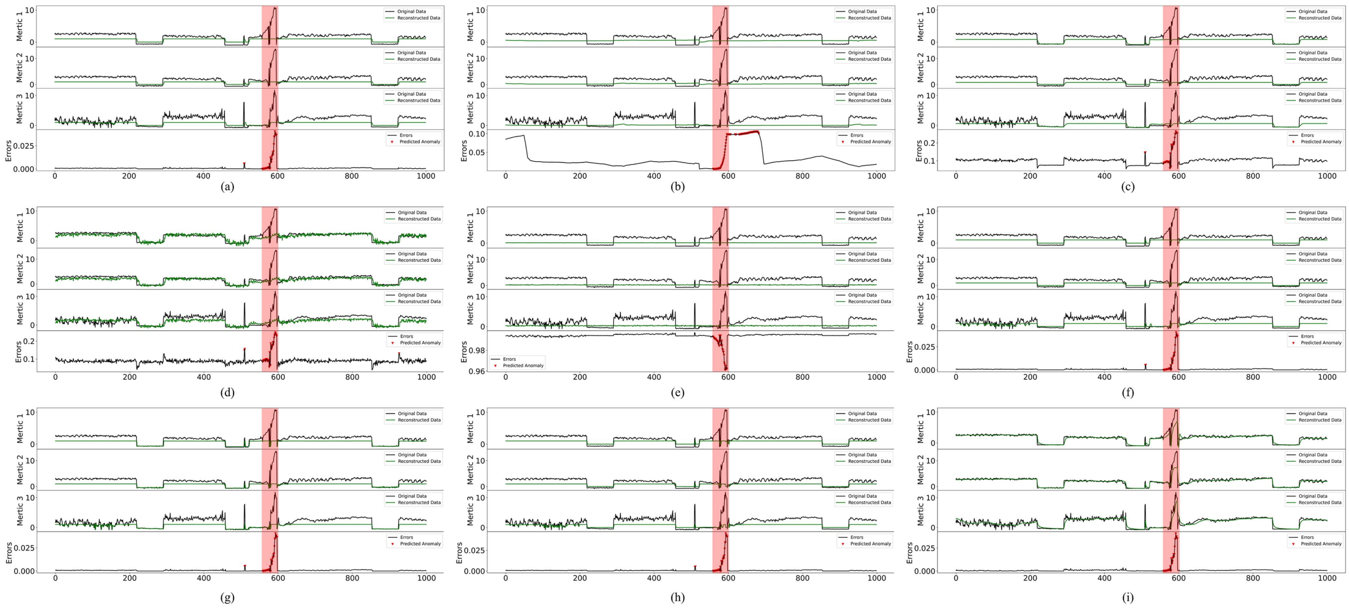

Fig. 5. Visualizations of model reconstruction and predicted anomalies. (a) DAGMM. (b) USAD. (c) EncDec-AD. (d) LSTM-VAE. (e) OmniAnomaly. (f) GDN. (g) MTAD-GAT. (h) TranAD. (i) ISRAE.

kernel high precision. After introducing noise into the MSPD, the performance of all models generally declines. Notably, the performance of GDN and MTAD-GAT, which use graph-based data correlations, drops significantly due to the noise's impact on graph construction. Models incorporating variational inference, such as LSTM-VAE and OmniAnomaly, show less performance degradation, as variational inference enhances robustness against noise. In contrast, the F1-score of EncDec-AE, which lacks variational inference, dropped by nearly 10%. Despite some improvement, USAD and MSCRED remain the worst-performing models, likely due to the difficulty of training them on this dataset.

In addition, to compare and analyze their ability to capture normal behavior patterns, we present the reconstructions of a pump generated by these models. The reconstruction errors and predicted anomalies are also shown to demonstrate the effectiveness of ISRAE compared to other SOTAs. These visualizations are illustrated in Fig. 5, where the red regions represent labeled anomalies. It is clear that LSTM-VAE and ISRAE have the best reconstruction abilities. Both models significantly underestimate anomalies, enhancing the effectiveness of reconstruction errors in differentiating between normal data and anomalies. OmniAnomaly performs the worst in reconstruction, confirming it has barely learned the normal behavior patterns of MTS. In addition, we observe false alarms by other models, while ISRAE does not, highlighting the advantage of ISRAE's inference kernel that fully considers the strong causal or logical relationships among the variables. Moreover, it is evident that the model effectively captures normal data trends while significantly underestimating anomalies. This occurs because stable trends in normal data incur smaller penalties from the differential constraint, whereas anomalies, due to their instability, incur larger penalties. These results demonstrate that the differential constraint enhances the model's anomaly detection capabilities.

We further consider different τ values of PA based on (36) to evaluate model performance comprehensively. Fig. 6 shows that our proposed ISLSTMAE and ISGRUAE do not exhibit a rapid decrease in performance across three metrics (precision, recall, F1-score) as τ increases, while other SOTAs deteriorate faster in at least two metrics. This indicates that our ISRAE anomaly detection algorithm has low sensitivity to PA, further demonstrating that the model can more accurately predict anomalies while the inference kernel can consistently reduce false alarms.

C. Model Discussion

1) Sensitivity Analysis of Hyperparameters: We discuss the model performance on the MSPD dataset by conducting sensitivity analysis. Based on (5), a hyperparameter α balances the tradeoff between the two terms in the loss function. Therefore, we analyze this hyperparameter to see how it affects ISRAE's performance. The results are illustrated in Fig. 7. Since the F1-score is the harmonic mean of precision and recall, we primarily focus on changes in the F1-score. It is evident that the F1-score of ISRAE shows a notable increase from $\alpha = 0$ to $\alpha = 0.01$, staying at a remarkably high level until $\alpha = 0.4$, where it starts to decrease. Given this consistently strong performance, we choose $\alpha = 0.11$ that maximizes the F1-score.

2) Ablation Study: We conduct an ablation study to demonstrate module's effectiveness in ISRAE. In addition to the two baselines (i.e., NAE and DAE), we also introduce a new baseline here that initializes the model without training. Comparing these baselines helps us understand whether our proposed model and its modules have positive effectiveness for anomaly detection. The results are presented in Table IV. The ISRAE model achieves optimal performance only when all three modules are applied simultaneously, with F1-scores of 0.8937 and 0.8870, respectively.

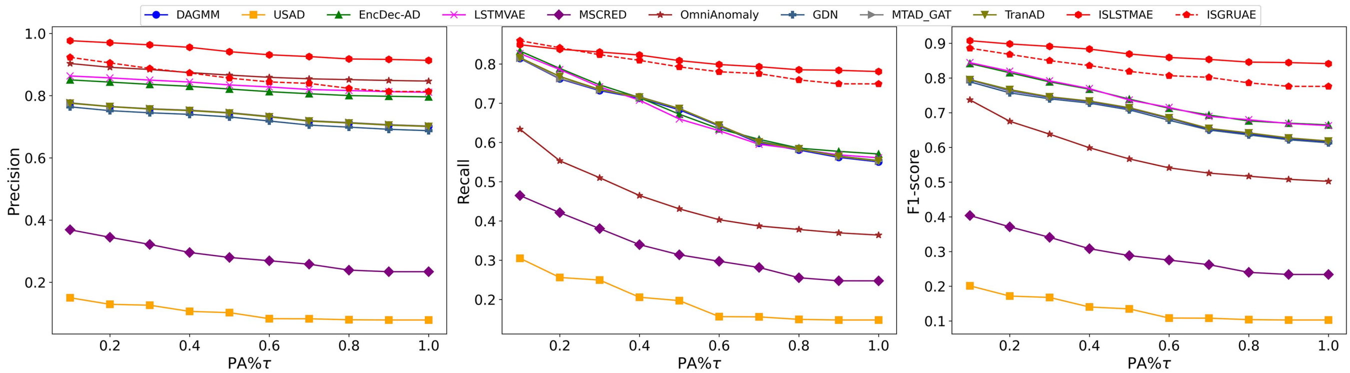

Fig. 6. Comparison results on the MSPD with different PA%τ.

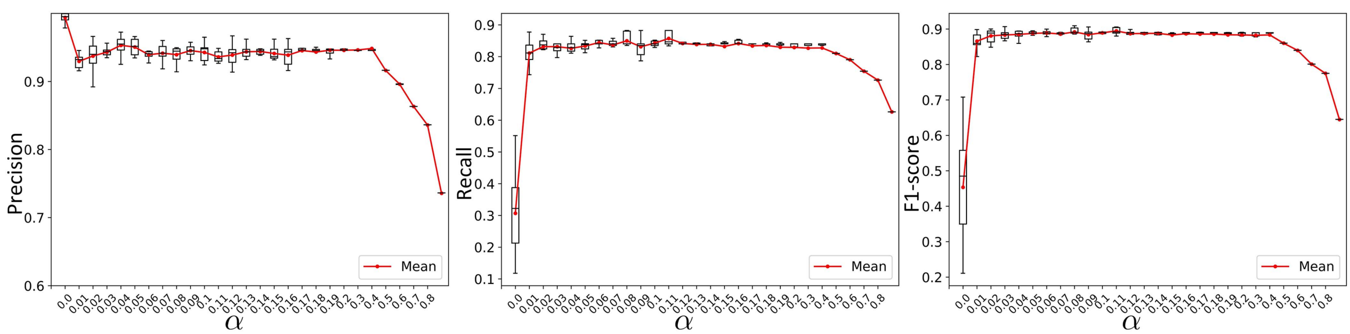

Fig. 7. Sensitivity analysis of α.

Table IV indicates that the effects of the different modules are interconnected rather than independent. The stacking module alone does not perform well but shows notable improvement when paired with either of the other two modules. This is because the stacking module can eliminate interference between variables during reconstruction, leading to better performance. However, without the other two modules, the model might achieve excessively high reconstruction performance, which could hinder effective anomaly identification. When only the differential constraint is introduced, without the other two modules, the F1-scores of the two models reach 0.8564 and 0.8301, respectively, which is very close to the baseline DAE. The differential constraint smooths the reconstruction results, increasing the difference between the reconstruction and the original data in unstable regions, thus enhancing performance. Similarly, introducing only the inference kernel results in an F1-score close to that of the baseline DAE, with good recall. The inference kernel can reduce false alarms for fake anomalies present in the private dataset. However, without the other two modules, the model's reconstruction capability remains significantly limited. Combining any two of the three modules above results in an F1-score of 0.85–0.87, significantly improving over the individual modules. This improvement is due to the complementary advantages of each module enhancing anomaly detection performance. For example, combining the inference kernel with either of the other two modules generally improves precision, as the inference kernel can better identify fake anomalies when reconstruction capabilities are enhanced. When all three modules are integrated, the performance of the ISRAE model reaches its optimum level.

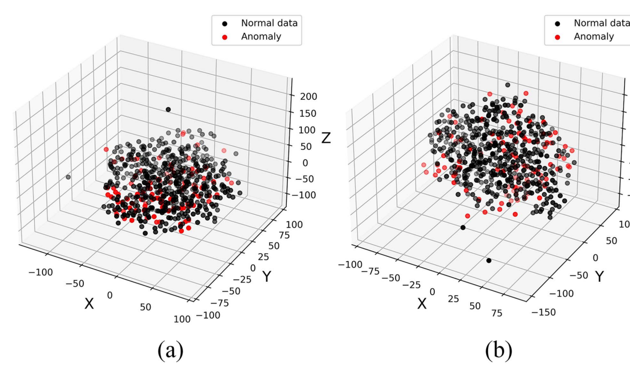

Fig. 8. Feature distribution of ISRAE. (a) ISLSTMAE. (b) ISGRUAE.

3) Distribution of ISRAE's Feature: Based on the main idea of ISRAE anomaly detection, we understand that when the reconstruction of anomalies is similar to that of normal data, greater reconstruction residuals make anomalies easier to detect. By visualizing the feature distribution of ISRAE in Fig. 8, we find that the features of anomalies and normal data are distributed similarly. This indicates that our proposed ISRAE effectively captures normal behavior patterns.

IV. Conclusion

In this article, we proposed the ISRAE model for anomaly detection in MTS under strong mechanistic contexts. ISRAE effectively mitigated interference among variables by using a stacking approach. The inference kernel, derived from specialist

TABLE IV
ABLATION STUDY RESULTS

Operation			Module			LSTMAE			GRUAE		
①	②	③	④	⑤	⑥	P	R	F1	P	R	F1
						.7842 (.0265)	.6954 (.1115)	.7371 (.0802)	.8362 (.0567)	.6853 (.0493)	.7533 (.0378)
			✓			.8099 (.0308)	.7483 (.1272)	.7711 (.0800)	.8881 (.0783)	.6177 (.1266)	.7175 (.0761)
				✓		.8891 (.0785)	.8260 (.0231)	.8564 (.0577)	.8620 (.1089)	.8004 (.0587)	.8301 (.1231)
					✓	.7932 (.0564)	.8763 (.1032)	.8327 (.0879)	.9391 (.0165)	.7733 (.0546)	.8482 (.0310)
			✓	✓		.8690 (.0274)	**.8833 (.0298)**	<u>.8753 (.0065)</u>	.8909 (.0231)	**.8576 (.0397)**	<u>.8739 (.0215)</u>
			✓		✓	.8534 (.0076)	<u>.8812 (.0689)</u>	.86701 (.0794)	**.9503 (.1168)**	.8035 (.0503)	.8698 (.0278)
				✓	✓	<u>.8959 (.0217)</u>	.8218 (.0124)	.8573 (.0102)	.9399 (.0105)	.8083 (.0229)	.8691 (.0105)
			✓	✓	✓	**.9391 (.0106)**	.8531 (.0256)	**.8937 (.0109)**	<u>.9405 (.0106)</u>	<u>.8401 (.0309)</u>	**.8870 (.0135)**
✓			✓	✓	✓	.2175 (.0509)	.4158 (.0812)	.2822 (.0379)	-	-	-
	✓		✓	✓	✓	.1024 (.0171)	.1429 (.0340)	.1179 (.0204)	.0577 (.0155)	.1150 (.0527)	.0678 (.0065)
		✓	✓	✓	✓	.8369 (.0000)	.8483 (.0000)	.8426 (.0000)	-	-	-

Note: (1) P, R, and F1 denote the metrics of precision, recall, and F1-score, respectively. (2) Boldface represents the best result in each column while the underline represents the suboptimal result. (3) ①-⑥ represent NAE; the model only initializes without training; DAE; the stacking module; the differential constraint in loss function; the inference kernel.

knowledge, enhanced anomaly detection accuracy and reduced false alarms. In addition, a new differential constraint in the loss function emphasized anomaly reconstruction errors by smoothing the reconstructions. Comparison results demonstrated that ISRAE achieved superior anomaly detection performance in MTS with strong mechanistic contexts. Reconstruction comparisons highlighted the model's ability to capture normal behavior patterns effectively. Following [27], we presented the PA results for all models at different τ values and introducee three baselines, confirming the superiority of our model. The ablation study confirmed the role of each module and revealed potential interaction effects among the key modules. In summary, ISRAE exceled at detecting anomalies in strong mechanistic contexts.

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